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(54) **APPARATUS FOR ANCHORING A VENTRICULAR ASSIST SYSTEM IN A BLOOD VESSEL, OPERATING METHOD, PRODUCTION METHOD FOR PRODUCING AN APPARATUS AND VENTRICULAR ASSIST SYSTEM**

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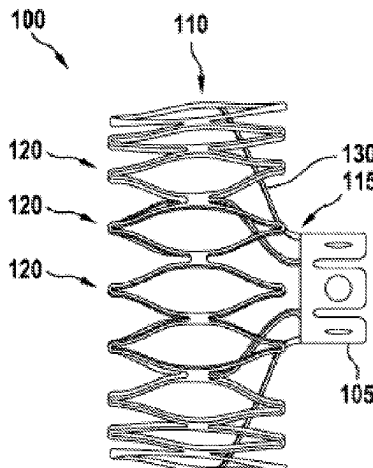
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(57)

ABSTRACT

The invention relates to an apparatus (100) for anchoring a ventricular assist system in a blood vessel, the apparatus (100) being able to assume an insertion state for insertion of the ventricular assist system into the blood vessel, and the apparatus (100) being able to assume an anchoring state in order to anchor the ventricular assist system in the blood vessel. The apparatus (100) has at least one fixing means (105) for fixing the apparatus (100) to the ventricular assist

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system (205), a crown (110) and a connection means (115). The crown (110) is formed from at least one unfolding element (120). The unfolding element (120) is designed to unfold during the transfer from the insertion state into the anchoring state in order to enlarge the diameter of the crown (110) so as to anchor the apparatus (100) in the blood vessel. The connection means (115) is designed to connect the crown (110) to the fixing means (105).

10 Claims, 16 Drawing Sheets

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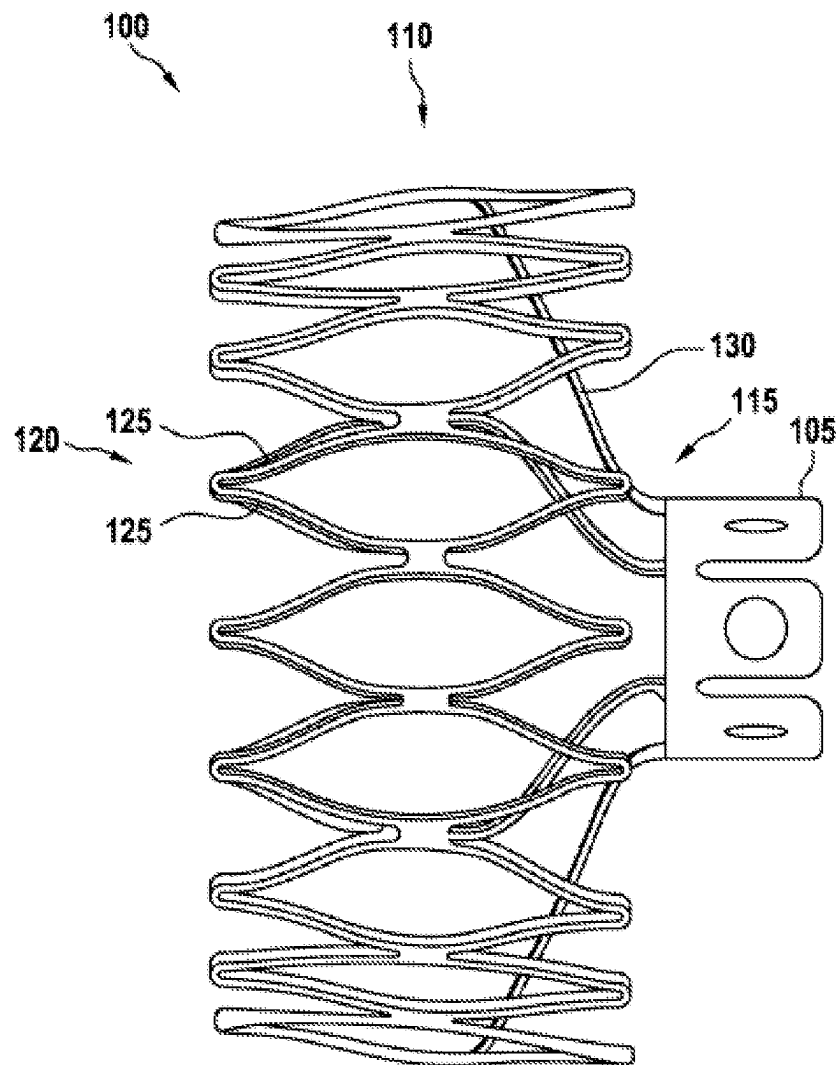
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Fig. 1



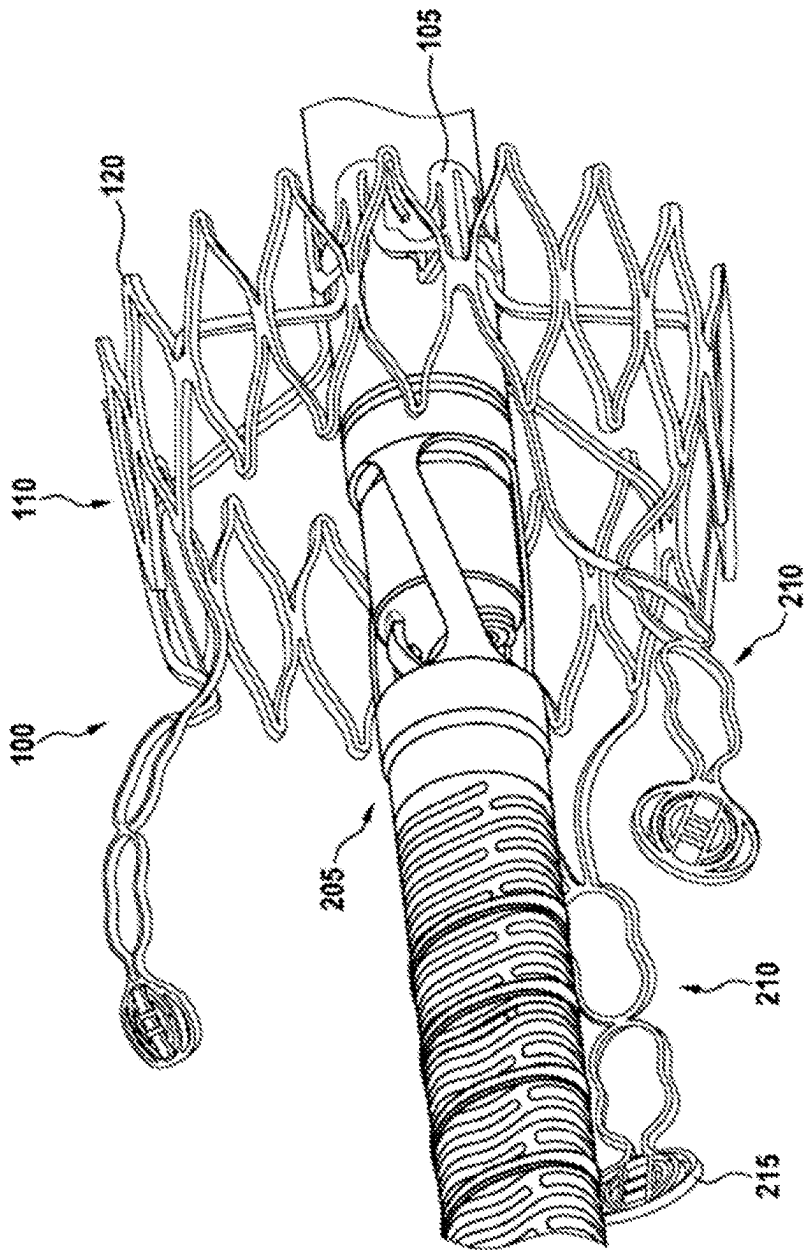


Fig. 2

Fig. 3

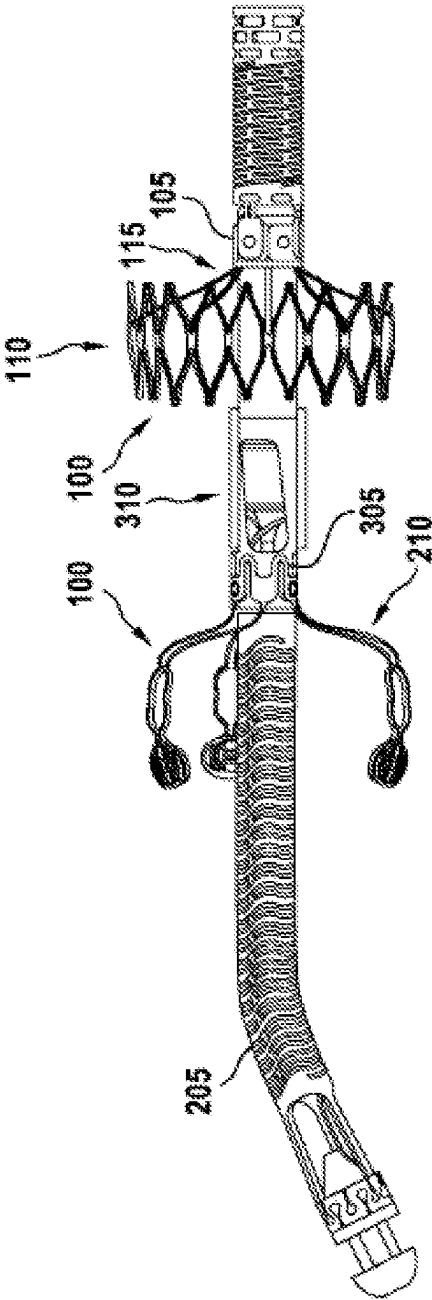


Fig. 4

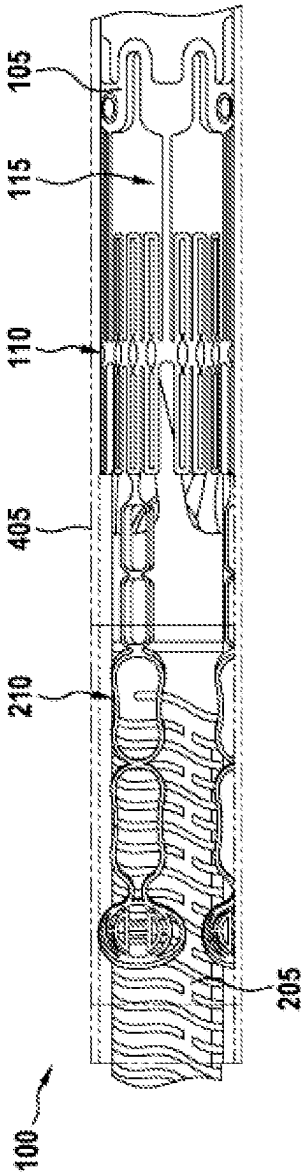


Fig. 5

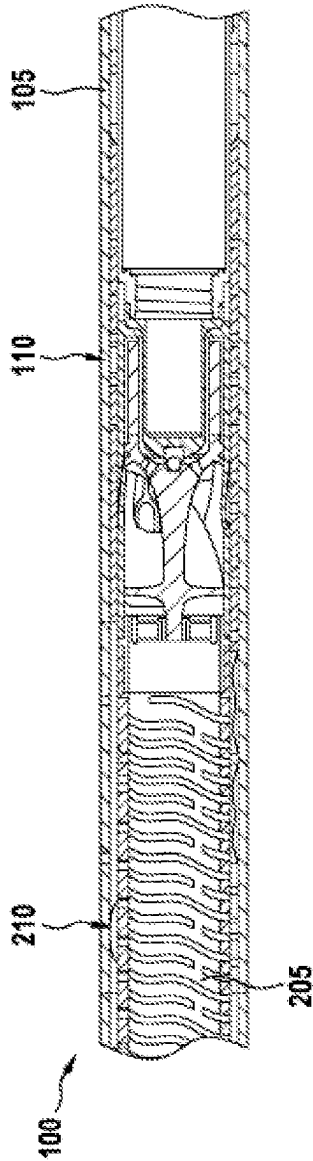


Fig. 6

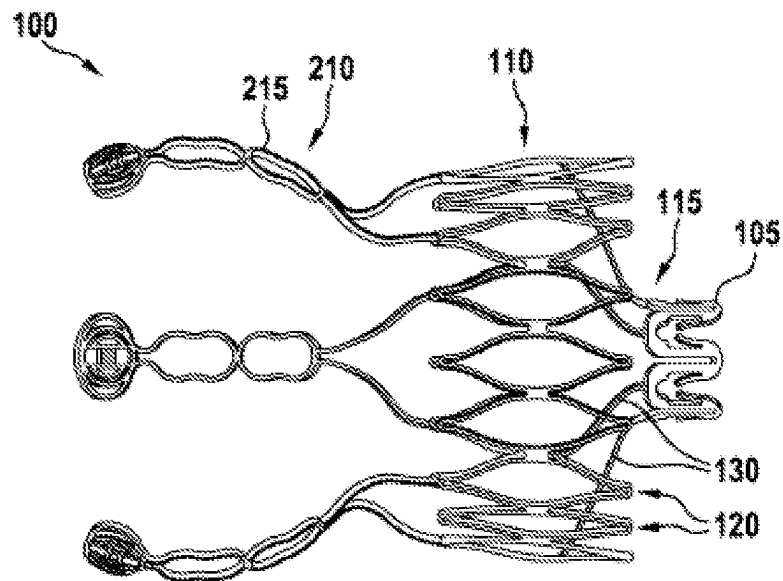


Fig. 7

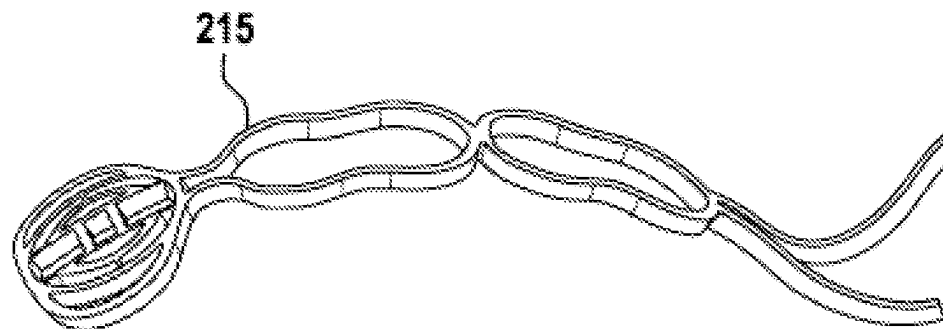


Fig. 8

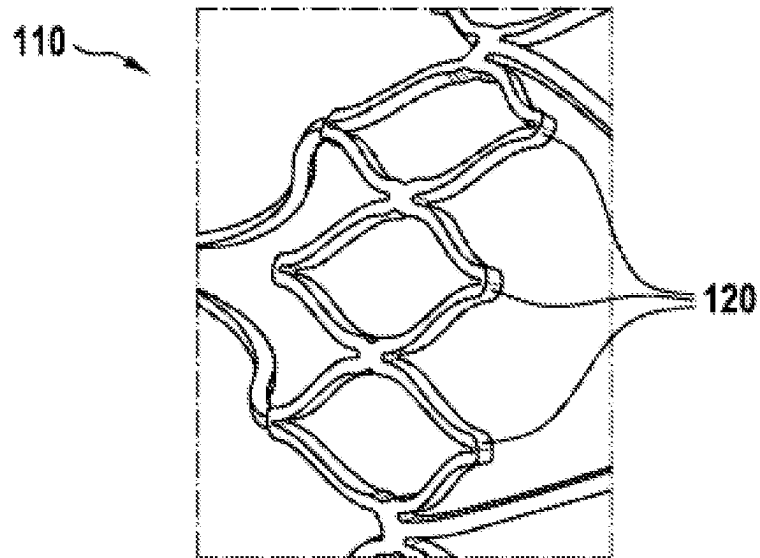


Fig. 9

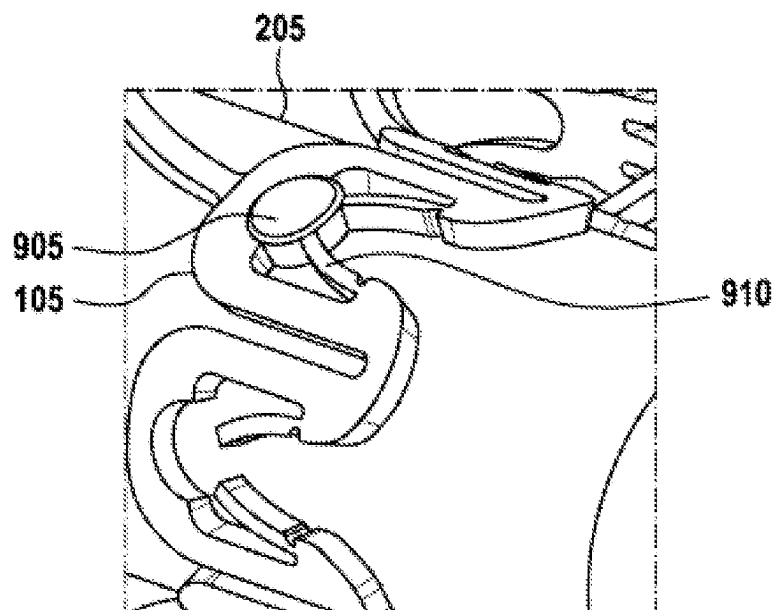


Fig. 10

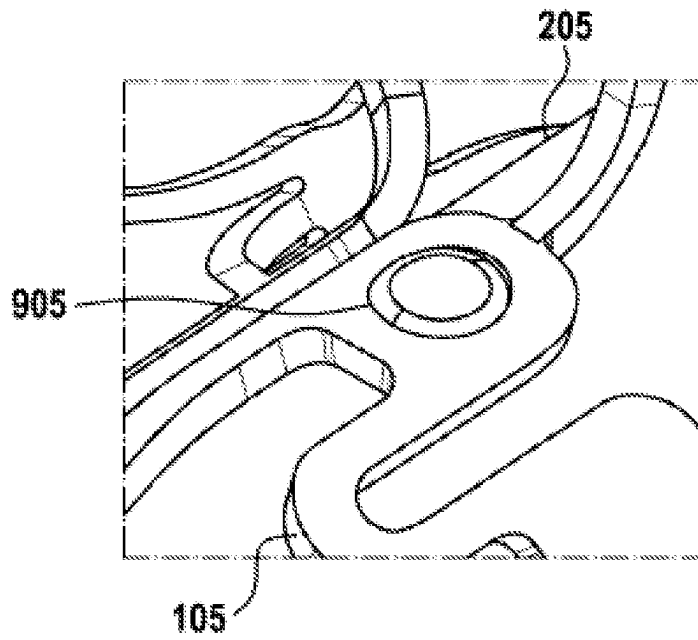


Fig. 11a

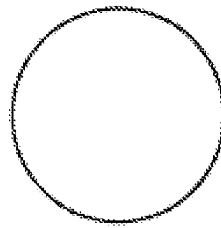


Fig. 11b

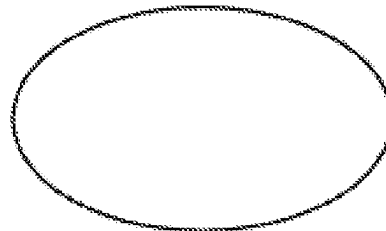


Fig. 11c

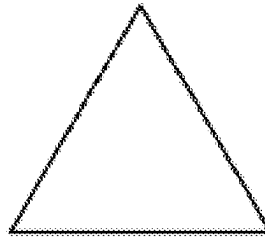


Fig. 11d

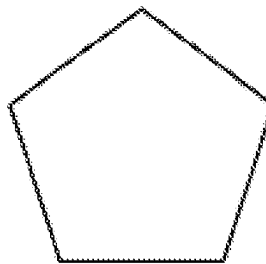


Fig. 11e



Fig. 11f

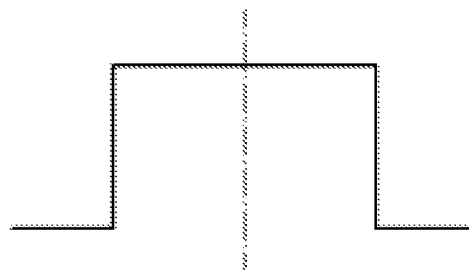


Fig. 11g

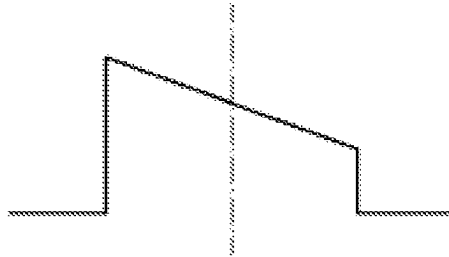


Fig. 11h

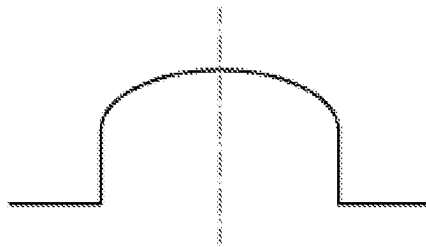


Fig. 11i

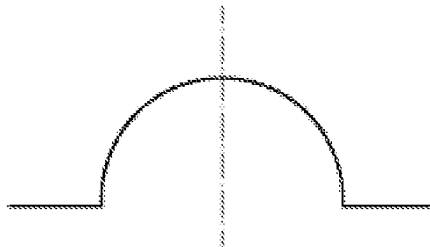


Fig. 11j

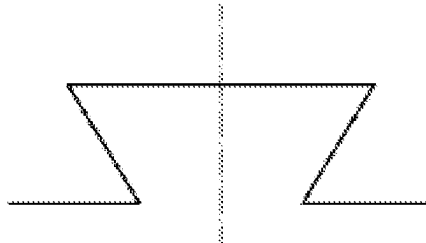


Fig. 12

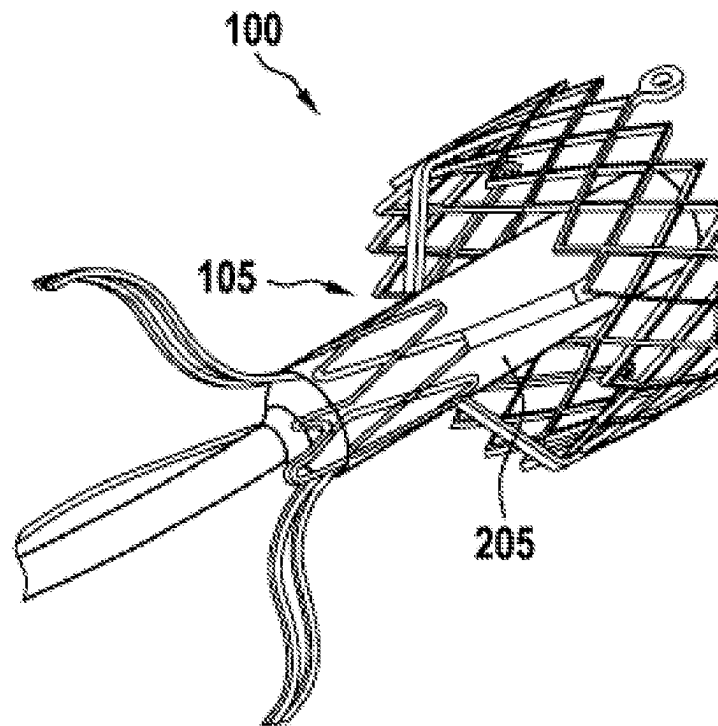


Fig. 13

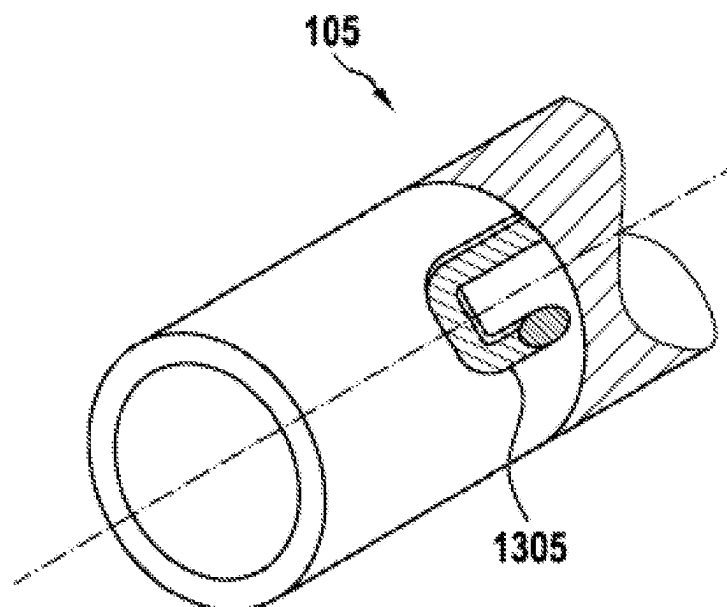


Fig. 14

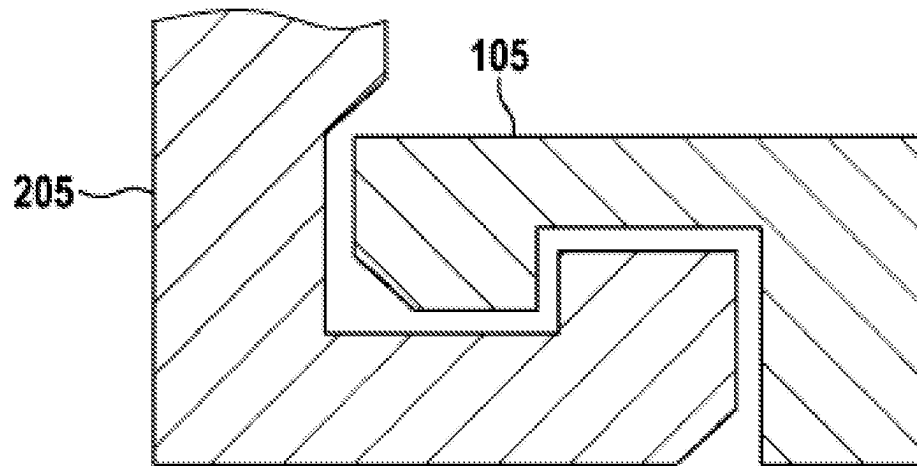


Fig. 15

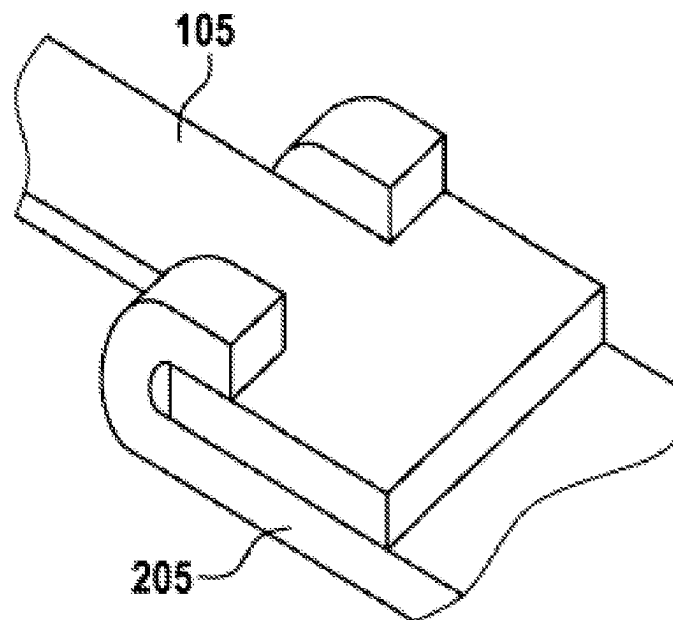


Fig. 16

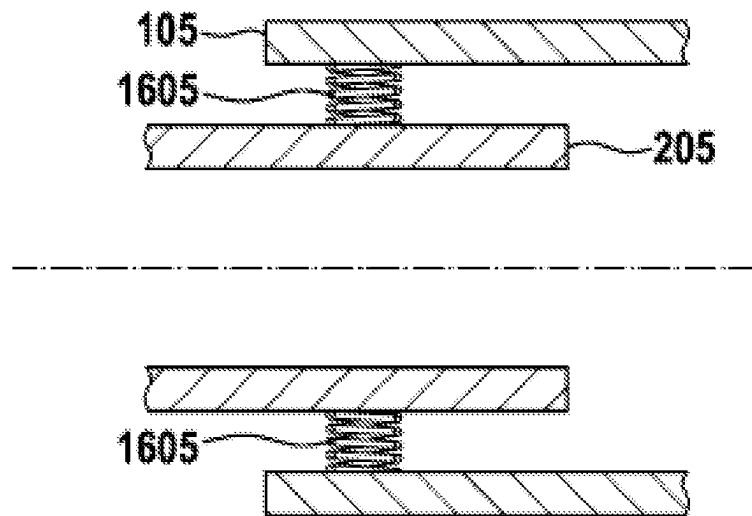


Fig. 17

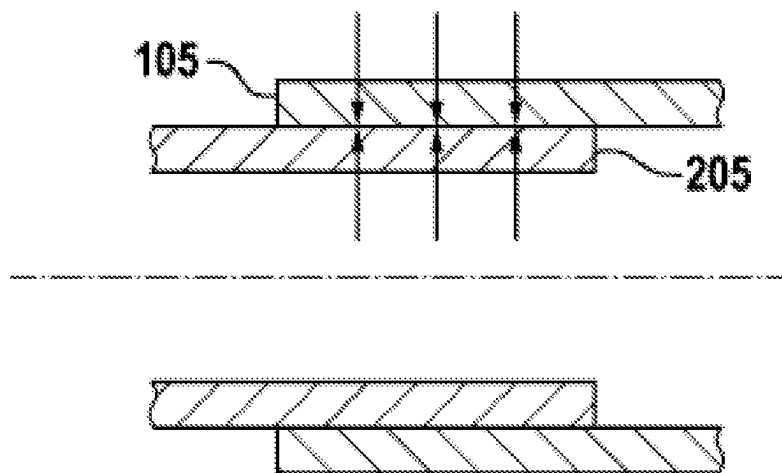


Fig. 18a

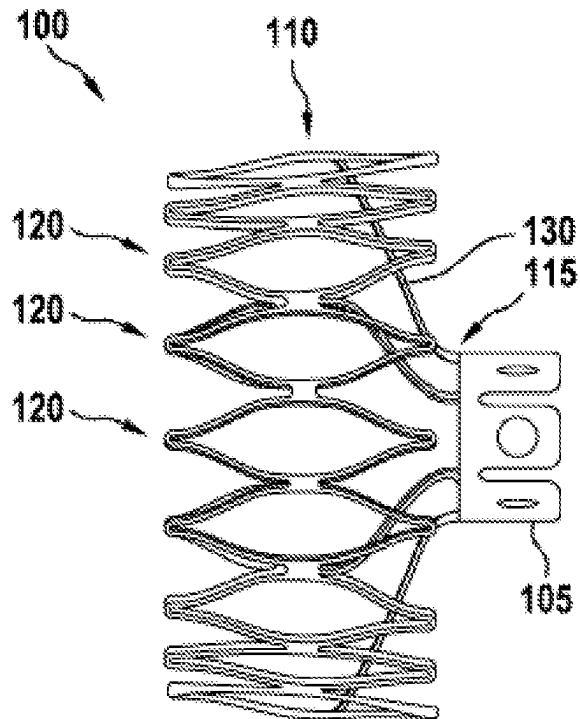


Fig. 18b

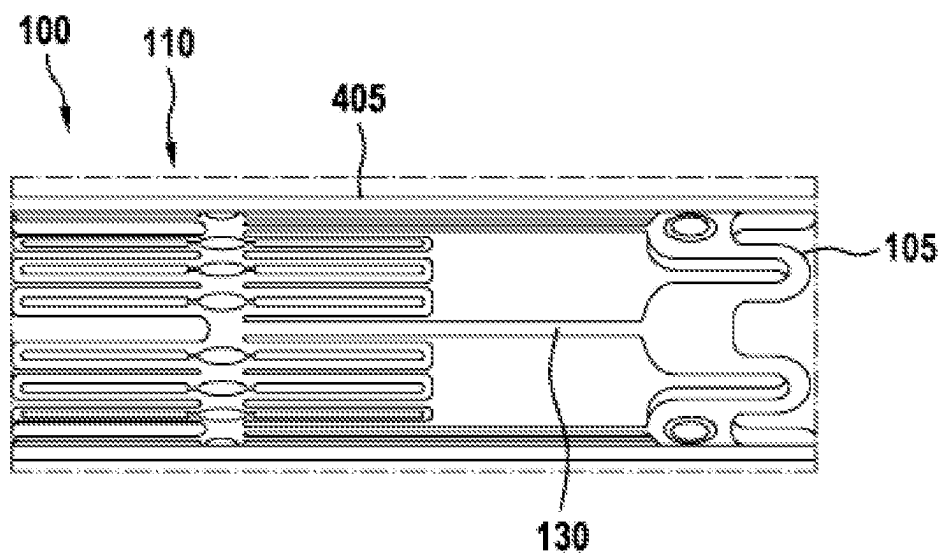


Fig. 18c

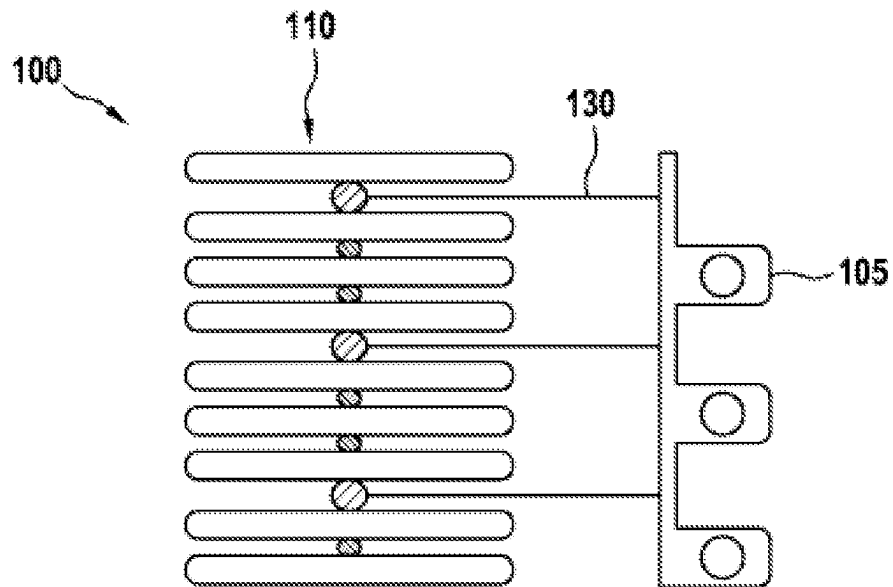


Fig. 19a

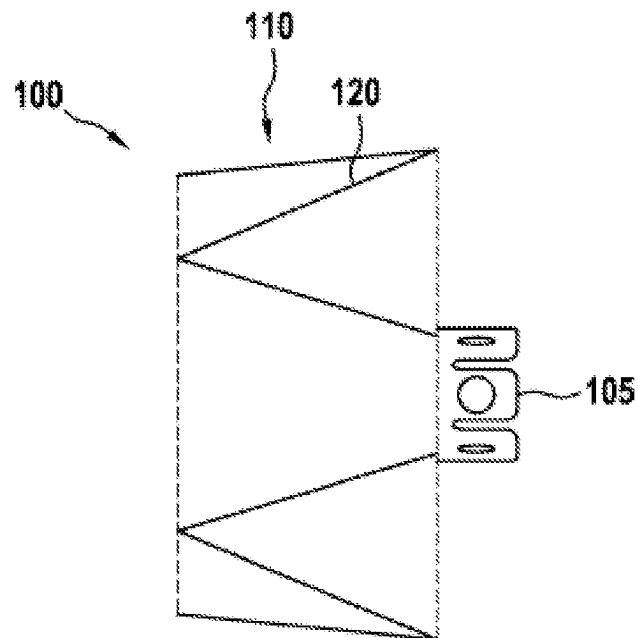


Fig. 19b

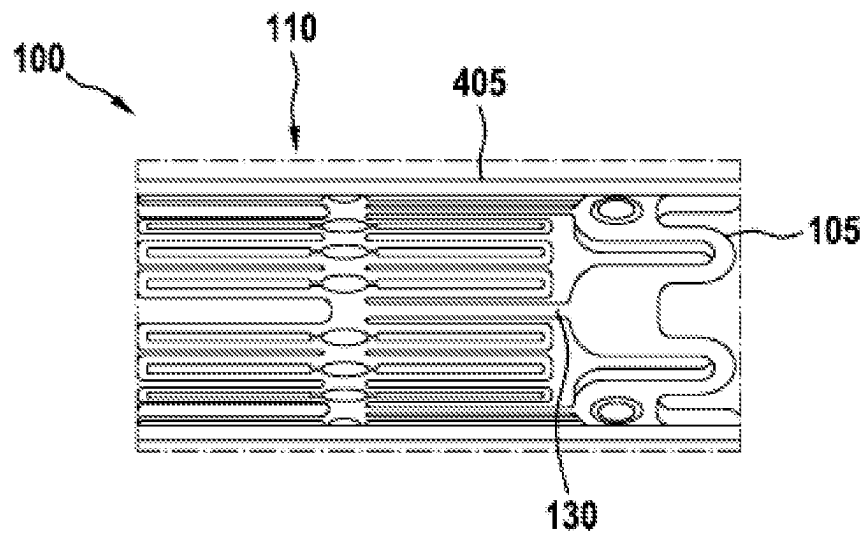


Fig. 19c

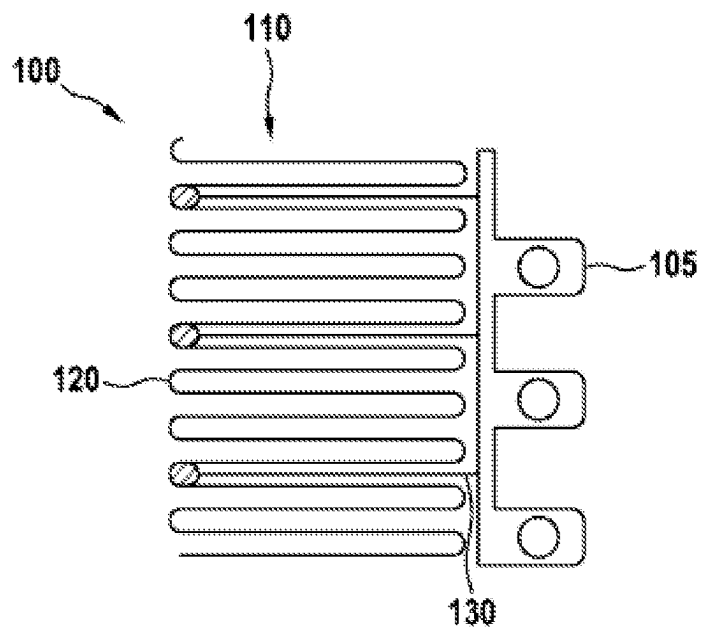


Fig. 20

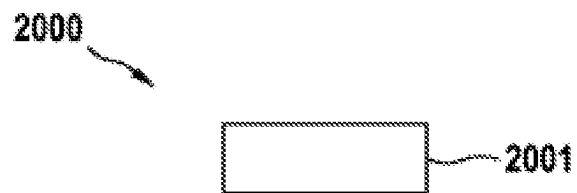
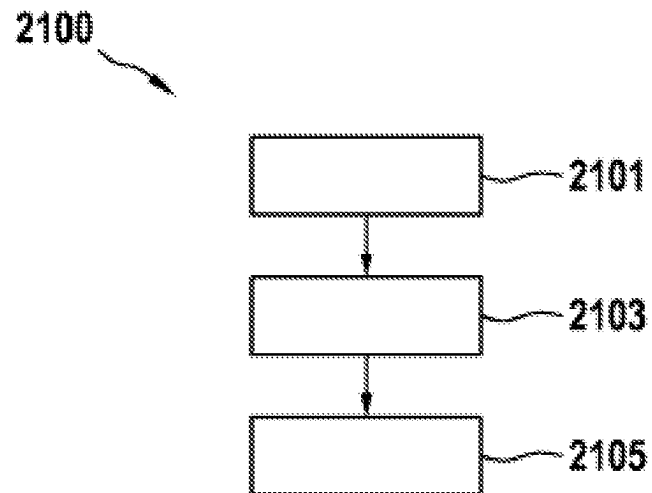


Fig. 21



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APPARATUS FOR ANCHORING A VENTRICULAR ASSIST SYSTEM IN A BLOOD VESSEL, OPERATING METHOD, PRODUCTION METHOD FOR PRODUCING AN APPARATUS AND VENTRICULAR ASSIST SYSTEM

BACKGROUND

Field

The invention relates to an apparatus for anchoring a cardiac support system in a blood vessel, a method for operating such an apparatus and a production method for producing such an apparatus and a cardiac support system.

Description of the Related Art

Cardiac support systems for long-term therapy are typically implanted via a complete or partial opening of the sternum, whereby a heart-lung machine can be used to enable extracorporeal blood circulation. This often involves punching a hole in the structural integrity of the myocardial tissue and the body's main artery, the aorta. Short-term intravascular cardiac support systems are delivered either percutaneously, i.e. through the skin, or surgically via various arterial accesses, for example via the femoral artery. The final positioning of the cardiac support systems can be visually supported intraoperatively, for example by means of ultrasound or radiological fluoroscopy. The implanted cardiac support systems have a high risk of dislocation, however, because there is no local fixing of the cardiac support system. The cardiac support system can therefore shift after implantation because it is not anchored at the implantation site, which can lead to malfunctioning of the pump or to an interruption of the therapy with the cardiac support system.

SUMMARY

The object of the invention is to create an apparatus that makes it possible to place a medical system, e.g. a cardiac support system, but also an implant, in particular in a blood vessel, e.g. inside the aorta, such that it has a generally constant spatial position relative to a section of the human or animal body, even over long periods of time, i.e. hours, days, weeks, months or, if necessary, even years.

This object is achieved by the apparatus for anchoring a cardiac support system in a blood vessel and the production method for an apparatus for anchoring a cardiac support system in a blood vessel, as described and claimed herein. Advantageous embodiments of the invention are specified in the dependent claims.

The apparatus for anchoring a cardiac support system in a blood vessel and a method for operating an apparatus for anchoring a cardiac support system as well as the production method for producing a corresponding apparatus and a cardiac support system having a corresponding apparatus according to the main claims are presented in the following. Advantageous further developments and improvements of the apparatus specified in the independent claim are possible using the measures listed in the dependent claims.

With the approach presented here, an implant, for example a cardiac support system, can be positioned and anchored in the blood vessel, in particular inside the aorta, by means of an apparatus with or without a spatial relationship to the aortic valve. For this purpose, the apparatus can be fixed to a heart pump. To be able to insert the cardiac

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support system in a minimally invasive manner, the apparatus can advantageously be folded, and the apparatus is designed such that it can unfold at the destination to position and anchor the cardiac support system. Using the apparatus, the cardiac support system can be aligned at the destination and placed in a targeted manner. The position of the cardiac support system advantageously remains unchanged in the long term due to the anchoring provided by the apparatus, as a result of which shifting or dislocation of the cardiac support system can be eliminated.

An apparatus for anchoring a cardiac support system in a blood vessel will be presented. The apparatus can assume an insertion state for inserting the cardiac support system into the blood vessel. The apparatus can also assume an anchoring state for anchoring the cardiac support system in the blood vessel. The apparatus comprises at least one fixing means for fixing the apparatus to the cardiac support system, a crown and a connection means, which is configured to connect the crown to the fixing means. The crown consists of at least one unfolding element. The unfolding element is designed to unfold during a transition from the insertion state into the anchoring state in order to increase the diameter of the crown to anchor the apparatus in the blood vessel.

The apparatus can be made of a biocompatible material to allow the apparatus to grow together with the blood vessel, for example the aorta, during long-term use of the apparatus. The apparatus can also be made of an elastic material that simultaneously exhibits a certain degree of rigidity, for example Nitinol. The cardiac support system can be a heart pump, for example, such as a right ventricular support system, a left ventricular support system, a biventricular support system or a vascular or valve prosthesis. The apparatus can furthermore also be used to position another component in a blood vessel and anchor it at its destination, for example a vascular or intracavitary implant such as a gastrointestinal, intrathecal, or intravesical implant. The insertion state can, for example, be understood to be the state assumed by the apparatus for insertion or during insertion of the cardiac support system into the blood vessel. The apparatus can be fixed to the heart pump, for example, and folded such that the apparatus can be inserted into a catheter along with the cardiac support system for minimally invasive insertion. The apparatus and the cardiac support system can be inserted in a minimally invasive manner through the leg artery, the femoral artery, for example. In the insertion state, the apparatus can correspondingly have a diameter that is less than the diameter of a human aorta. The anchoring state can be understood to be a state in which, after being inserted and aligned at the destination, the apparatus is unfolded to increase the diameter of the crown in order to anchor the cardiac support system in a force-locking manner at the destination and thus advantageously prevent shifting or dislocation of the cardiac support system. In the anchoring state, i.e. in the unfolded state, the apparatus or a component of the apparatus, for example the crown, can have an inner diameter that is slightly larger than the inner diameter of the blood vessel in which it is anchored, for example a total inner diameter in the range of 20-30 mm, for example 23 mm, for anchoring to or behind a human aortic valve. In the anchoring state, the outer contour of at least one part of the apparatus, for example the crown, can furthermore have a shape that corresponds to the not exactly circular aortic anatomy of a human aorta. The apparatus can thus be held at the destination in the anchoring state via a radial frictional connection.

The fixing means can be configured to fix the apparatus to an implant, for example the cardiac support system. For this

purpose, the fixing means can comprise at least one fixing element for creating a form-locking and/or force-locking connection to a counterfixing element disposed on the implant. The crown can be shaped like a ring. Depending on the embodiment, one single unfolding element can be shaped like a ring, or a plurality of unfolding elements can be strung together in a ring-like manner. Unfolding allows the at least one unfolding element to expand. The connection means can comprise at least one elongated, for example wire-shaped, strut, which is fastened to both the crown and the fixing means. The connection means can be designed to be flexible in order to allow the diameter of the crown to be increased.

According to one advantageous embodiment, at least the unfolding element of the crown of the apparatus can be made of shape memory material. The unfolding element can be made of a biocompatible shape memory polymer, for example, or a biocompatible shape memory alloy, such as Nitinol. It is furthermore also possible for the entire apparatus to be made of a shape memory alloy, for example Nitinol. Due to its shape memory properties, the use of a shape memory material such as Nitinol enables a particularly elegant and simple realization of the insertion state and the unfolding of the unfolding element during the transition into the anchoring state. The use of Nitinol as a shape memory material is advantageous, because the Nitinol material is a proven material in the field of medicine, in particular in the field of cardiovascular medicine, for example for heart valve prostheses, stents and vascular prostheses, due to its biocompatibility and the shape memory property, which makes it possible to deliver and place even complex structures, like the apparatus presented here, in a small installation space at the destination.

According to one embodiment, the apparatus can also comprise an arching device having at least one foot. The arching device is designed to unfold during the transition from the insertion state into the anchoring state to position the at least one foot in the blood vessel. By positioning the at least one foot in the blood vessel, the apparatus can be aligned and positioned in the blood vessel, for example, before it is anchored. The arching device can be connected to the crown. Alternatively, the arching device can comprise an arch fixing device for fixing the arching device to the cardiac support system. The apparatus can therefore be configured in one piece if the fixing means, the crown, the connection means and the arching device are coupled to one another, or the apparatus can be in two parts if the arching device is fixed to the cardiac support system by means of its own arch fixing device, and the other components of the apparatus are coupled to one another. The one-piece embodiment can be advantageous with regard to the folding of the apparatus for the insertion state; the two-piece embodiment can be advantageous depending on the design of the cardiac support system. If, for example, the cardiac support system comprises a pump with a motor, a two-part embodiment may provide more flexibility with respect to the installation space of the cardiac support system, for example by allowing the arching device to be fixed to one end of a motor-coupling-pump unit of the cardiac support system and the second part of the apparatus to be fixed to the other end of the motor-coupling-pump unit of the cardiac support system. By means of the arching device, the apparatus can advantageously be aligned and positioned at the destination via the unfolding of the at least one foot. The foot can have an atraumatic shape, for example, so as not to injure the blood vessel during unfolding and in the anchoring state. The arching device can advantageously be designed to enable alignment of the

apparatus for the anchoring state when positioning the at least one foot by the configuration of the arching device and the at least one foot.

According to a further embodiment, the arching device can comprise three feet, in particular wherein, for positioning the feet, said feet are formed in a respective cusp of a heart valve. This embodiment is advantageous in terms of being able to position the apparatus and with it the cardiac support system particularly precisely, in particular when the cardiac support system is positioned and anchored behind the aortic valve, for example, by means of the apparatus. The three feet can have a shape adapted to a peanut shape of the cusps, for example, in order to advantageously achieve a particularly advantageous balance between contact surface and torsional rigidity of the three feet.

For a catheter-supported minimally invasive implantation of the cardiac support system connected to the apparatus, it is advantageous if the apparatus is cylindrical in the insertion state as according to one embodiment. For this purpose, the apparatus can, for example be folded. If the apparatus is made of Nitinol, for example, the apparatus can be cut out of a tube and a shape corresponding to the anchoring state can then be embossed by means of a heat treatment. For the insertion state, the apparatus can be folded to correspond to the original cylindrical tube geometry.

It can also be advantageous if the unfolding element has an inclined position in the anchoring state as according to one embodiment. The inclined position can be understood to be a specific angle of the unfolding element relative to the longitudinal axis of the apparatus. In the anchoring state, the unfolding element can be inclined at an angle to support a force-locking connection between the crown and the blood vessel in the anchoring state. The angle relative to the longitudinal axis of the apparatus can be between 20° and 30°, for example, in particular 25°. The inclined position of the unfolding element can advantageously increase the pressing force of the crown on the blood vessel.

According to one embodiment, the crown can comprise a plurality of unfolding elements coupled to one another. Each of the unfolding elements can comprise two unfolding rods connected at their ends and the distance between the two unfolding rods can be smaller in the insertion state than in the anchoring state. The unfolding elements can be coupled to one another by means of a connection to one of the unfolding rods of an adjacent unfolding element. The two unfolding rods can be connected to one another such that, in the anchoring state of the apparatus, each unfolding element forms a rhomb shape with rounded corners in axial direction to the longitudinal axis of the apparatus. Such a configuration of a plurality of rhomb-shaped unfolding elements coupled to one another can correspond to a standard cross-section of a vascular stent, which can be advantageous when producing the apparatus.

According to one embodiment the unfolding element can alternatively comprise a plurality of loops arranged in a meandering manner, wherein the distance between the loops is smaller in the insertion state than in the anchoring state. This embodiment provides a particularly space-saving folding of the apparatus for the insertion state.

According to one embodiment, the connection means can comprise at least one flexure strut. The flexure strut is designed to open during the transition from the insertion state into the anchoring state to allow the crown to unfold. The flexure strut enables a particularly elegant and space-saving connection between the fixing means and the crown. The flexure strut can be made of, for example, an elastic material, for example also Nitinol.

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According to a further embodiment, a first end of the flexure strut can furthermore be fastened to the fixing means. A second end of the flexure strut can be fastened to a connection between two adjacent unfolding elements, or the second end of the flexure strut can be fastened to an end of the crown facing away from the fixing means. The flexure strut can thus be configured to correspond to the shape of the unfolding element of the crown to enable efficient unfolding of the crown depending on the embodiment and, depending on the embodiment of the unfolding element, enable compact folding with respect to the axial length of the apparatus for the insertion state.

According to one embodiment, the fixing means can be designed to fix the apparatus in a form-locking and/or force-locking manner to the cardiac support system. The apparatus can thus be connected to the cardiac support system in a stable manner in order to absorb forces that occur, for example, during implantation or during the operating time of the cardiac support system. For this purpose, the fixing means can comprise an element for form-locking engagement, for example, or a recess for receiving an element in a form-locking manner that is disposed on the housing of the cardiac support system, for example. The element for form-locking fixing can have different cross-sections, for example, and the element or the corresponding recess can, for example, be round, oval, triangular, polygonal or star-shaped. For this purpose, both the fixing means and the cardiac support system, for example the housing of the cardiac support system, can comprise elements that enable the fixing means to be snapped into or anchored to the cardiac support system. Additionally or alternatively, the fixing means can, for example, comprise a bayonet connection or a clip connection or a hook. Additionally or alternatively, the fixing means can also be implemented by means of a material-locking connection. The fixing means can, for example, be configured to enable a rotational movement between the apparatus and the cardiac support system.

According to one embodiment, the apparatus can also comprise a sleeve. The sleeve can be movable relative to the crown. The sleeve can furthermore be designed to enclose at least the crown in the insertion state and to release the crown to initiate the transition into the anchoring state. If, as according to one embodiment, the apparatus comprises the arching device, the sleeve can also be movable relative to the arching device and can furthermore be designed to also enclose the arching device in the insertion state and to release the arching device to initiate the transition into the anchoring state. The sleeve can also be designed to enclose and then release all of the other components of the apparatus. The sleeve can be cylindrically shaped, for example, and designed such that the apparatus with the sleeve can be inserted into a commercially available catheter in the insertion state. The sleeve can, for example, advantageously be used to hold down the other components of the apparatus in the folded state of the apparatus and thereby additionally stabilize them in the insertion state; for example also when components of the apparatus or the entire apparatus are made of a shape memory material. The sleeve can furthermore be removed gradually during the transition from the insertion state into the anchoring state, for example via a controlled mechanism that can be controlled electrically, for example, or by manually pulling back the sleeve. The apparatus can thus, for example, be unfolded incrementally to advantageously unfold the apparatus in a controlled manner and position it prior to anchoring, or the sleeve can be moved forward again to realign the apparatus or correct the positioning of the apparatus.

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A method for operating the apparatus according to one embodiment is also presented along with this approach. The method comprises at least one unfolding step. In the unfolding step, the unfolding element of the crown of the apparatus is unfolded during the transition from the insertion state into the anchoring state to increase the diameter of the crown. The unfolding step can, for example, also be carried out to increase the diameter of the crown to anchor the apparatus in the blood vessel. The method can be carried out when the apparatus is disposed inside a blood vessel and also when the apparatus is located outside a blood vessel, for example in order to connect the apparatus to a component of an implant by increasing the diameter of the crown.

A production method for producing an apparatus for anchoring a cardiac support system in a blood vessel is presented as well. The apparatus can assume an insertion state for inserting the cardiac support system into the blood vessel, and the apparatus can furthermore assume an anchoring state for anchoring the cardiac support system in the blood vessel. The production method comprises at least one step for providing, one step for forming and one step for heat treating. In the providing step, a semi-finished product made of a shape memory material is provided. In the forming step, a fixing means for fixing the apparatus to the cardiac support system is formed. The forming step also includes forming a crown consisting of at least one unfolding element, wherein the unfolding element is designed to unfold during a transition from the insertion state into the anchoring state in order to increase the diameter of the crown to anchor the apparatus in the blood vessel. Furthermore, in the forming step, a connection means is formed to connect the crown to the connection means. The fixing means, the crown and the connection means are formed from the semi-finished product. In the heat-treating step, the fixing means, the crown and the connection means are heat-treated to emboss the shape of the anchoring state.

A cardiac support system having an apparatus according to an embodiment is presented as well. The apparatus can be fixed to the cardiac support system, for example. The cardiac support system can, for example, comprise a heart pump with a motor-coupling-pump unit. The cardiac support system can furthermore comprise a housing, which is designed to be connected to the apparatus in a form-locking and/or force-locking manner, for example by means of the fixing means of the apparatus. According to this embodiment, the size of the cardiac support system and/or the dimensions of the apparatus can advantageously be selected or changed in a patient-specific manner.

BRIEF DESCRIPTION OF THE DRAWINGS

Design examples of the approach presented here are shown in the drawings and explained in more detail in the following description. The figures show:

FIG. 1 a schematic illustration of an apparatus for anchoring a cardiac support system in a blood vessel according to a design example;

FIG. 2 a schematic illustration of a part of a cardiac support system with an apparatus for anchoring a cardiac support system in a blood vessel according to a design example;

FIG. 3 a schematic illustration of a cardiac support system with an apparatus for anchoring a cardiac support system in a blood vessel according to a design example;

FIGS. 4 and 5 a schematic illustration of an insertion state of an apparatus for anchoring a cardiac support system in a blood vessel according to a design example;

FIG. 6 a schematic illustration of an apparatus for anchoring a cardiac support system in a blood vessel having an arching device according to a design example;

FIG. 7 a schematic illustration of a foot of an apparatus for anchoring a cardiac support system in a blood vessel according to a design example;

FIG. 8 a schematic illustration of a part of a crown of an apparatus for anchoring a cardiac support system in a blood vessel according to a design example;

FIGS. 9 and 10 a schematic illustration of a fixing means of an apparatus for anchoring a cardiac support system in a blood vessel according to a design example;

FIGS. 11a to 11j a schematic illustration of a design of a form-locking fixing of an apparatus for anchoring a cardiac support system in a blood vessel according to a design example;

FIG. 12 a schematic illustration of a force-locking fixing of an apparatus for anchoring a cardiac support system in a blood vessel to a cardiac support system according to a design example;

FIG. 13 to 17 a schematic illustration of a part of a fixing means of an apparatus for anchoring a cardiac support system in a blood vessel according to a design example;

FIG. 18a to 18c a schematic illustration of an apparatus for anchoring a cardiac support system in a blood vessel according to a design example;

FIG. 19a to 19c a schematic illustration of an apparatus for anchoring a cardiac support system in a blood vessel according to a design example;

FIG. 20 a flow diagram of a method for operating an apparatus for anchoring a cardiac support system in a blood vessel according to a design example; and

FIG. 21 a flow diagram of a production method for producing an apparatus for anchoring a cardiac support system in a blood vessel according to a design example.

DETAILED DESCRIPTION

In the following description of favorable design examples of the present invention, the same or similar reference signs are used for the elements shown in the various figures, which have a similar effect, whereby a repeated description of these elements is omitted.

FIG. 1 shows a schematic illustration of an apparatus 100 for anchoring a cardiac support system in a blood vessel according to a design example. Even if the apparatus 100 is described here and in the following in the context of a cardiac support system, the apparatus 100 can also be used to anchor other implants in a blood vessel.

The figure shows a side view of the unfolded apparatus 100. The apparatus 100 can assume an insertion state for inserting the cardiac support system into the blood vessel. The apparatus 100 can also assume the anchoring state shown in FIG. 1 to anchor the cardiac support system in the blood vessel.

The apparatus 100 comprises at least one fixing means 105 for fixing the apparatus 100 to the cardiac support system, a crown 110 and a connection means 115. The crown 110 consists of at least one unfolding element 120. The unfolding element 120 is designed to unfold during a transition from the insertion state into the anchoring state in order to increase the diameter of the crown 110 to anchor the apparatus 100 in the blood vessel. According to the shown design example, the crown 110 comprises a plurality of unfolding elements 120. The connection means 115 is designed to connect the crown 110 to the fixing means 105.

The apparatus 100 can advantageously be used to prevent the cardiac support system from shifting or moving, because the crown 110 can be used to anchor the apparatus at the implantation site. The apparatus 100 can furthermore enable a defined positioning of the cardiac support system, which is also mentioned in the following. The apparatus can be made at least partially of a shape memory material such as Nitinol. In FIG. 1 shown here, the apparatus 100 is configured as a Nitinol frame for anchoring and positioning a cardiac support system in a blood vessel. The apparatus 100 can have a design for long-term use as well as a retrievable design for short-term implantations, for example via a controlled transpositioning of the implanted and anchored apparatus 100 from the anchoring state into the insertion state.

The crown 110 and the fixing means 105 of the apparatus 100 extend along a longitudinal axis of the apparatus 100, which can correspond to the axis of a catheter in which the cardiac support system is inserted via the leg artery in a minimally invasive manner.

In the illustration shown here, the unfolding element 120 and with it the crown 110 is unfolded in accordance with the anchoring state. According to the shown design example, the unfolding element 120 can have an inclined position relative to the longitudinal axis of the apparatus 100 in the anchoring state. The unfolding element 120 can be inclined at a specific angle, for example a 250 angle, in order to produce an increased pressing force of the crown 110 for the force-locking anchoring of the apparatus 100 in the blood vessel.

As in the design example shown here, the crown 110 can comprise a plurality of unfolding elements 120 coupled to one another, wherein each of the unfolding elements 120 comprises two unfolding rods 125 connected at their ends. The distance between central sections of the two unfolding rods 125 of each unfolding element 120 is smaller in the insertion state than in the anchoring state. According to this design example, the central sections of all unfolding elements 120 are arranged on a circular path. In the unfolded state, each unfolding element 120 can have a rhomb shape with rounded corners, wherein the rhomb shape is configured by removing the two unfolding rods 125 connected at their ends. The unfolding elements 120 coupled to each other form a grid-like ring. The unfolding elements 120 can be identically shaped.

The connection means 115 optionally comprises at least one flexure strut 130. The flexure strut 130 is designed to open during the transition from the insertion state into the anchoring state to allow the crown 110 to unfold. The connection means 115 can also comprise a plurality of flexure struts 130, for example to enable a particularly uniform unfolding of the unfolding element 120 and thus of the crown 110. As an example, in the design example shown here, the apparatus 100 comprises four equally spaced flexure struts 130.

According to a design example, a first end of the flexure strut 130 is fastened to the fixing means 105 and, as shown here, a second end of the flexure strut 130 is fastened to a connection between two adjacent unfolding elements 120. The second end of the flexure strut 130 can alternatively be fastened to an end of the crown 110 facing away from the fixing means 105.

According to a design example, the fixing means 105 is designed to fix the apparatus 100 in a form-locking and/or force-locking manner to the cardiac support system. For this purpose, the fixing means 105 can, for example as shown here, comprise a recess for receiving a disposing element in

a form-locking manner, wherein the corresponding element can be formed in a component of the cardiac support system to be fixed.

According to a design example, the fixing means **105** is shaped as a ring comprising a plurality of recesses. In the unfolded state of the crown **110**, the ring-shaped crown **110** has a larger diameter than the ring-shaped fixing means **105**.

According to a design example, the crown **110** and the fixing means **105** do not overlap or overlap only slightly.

According to a design example, a longitudinal axis of the apparatus **100** extends centrally through the crown **110** and the fixing means **105**. According to a design example, the unfolding elements **120** have an inclined position in the unfolded state, wherein the ends of the unfolding elements **120** facing the fixing means **105** are further away from the longitudinal axis of the apparatus **100** than the ends of the unfolding elements **120** facing away from the fixing means **105**.

FIG. 2 shows a schematic illustration of a part of a cardiac support system **205** with an apparatus **100** for anchoring the cardiac support system **205** in a blood vessel according to a design example. A section of a heart pump having tube-shaped components, which include a motor and an impeller of the heart pump, is shown as an example of a part of the cardiac support system **205**. In the plan view shown here, the apparatus **100** is fixed to a housing section of the cardiac support system **205**.

According to a design example, the unfolding element **120** can be made of a shape memory material. It is also possible for multiple components, or the apparatus **100** as a whole, to be made of a shape memory material, for example Nitinol as shown here.

According to the design example shown here, the apparatus **100** comprises an arching device **210** having at least one foot **215**. The arching device **210** is designed to unfold during the transition from the insertion state into the anchoring state and thus enable the positioning of the at least one foot **215** in the blood vessel. The arching device **210** here is connected to the crown **110**. The arching device **210** can alternatively also comprise an arch fixing device for fixing the arching device **210** to the cardiac support system **205**; this is shown in FIG. 3 below.

The arching device **210** can comprise three feet **215**, as shown here. To position the feet **215**, the feet **215** can in particular be formed in a cusp of a heart valve, for example when the cardiac support system **205** fixed by the apparatus **100** is positioned and anchored inside a human aorta directly behind an aortic valve.

The fixing means **105** can fix the apparatus **100** to the cardiac support system by means of a form-locking, force-locking or material-locking connection mechanism. For this purpose, as shown here for example, the cardiac support system **205** can comprise a connection element for form-locking engagement of the fixing means **105**, and the fixing means **105** can comprise a corresponding material recess for engagement or a correspondingly formed connection element for engagement, such as the clip connection shown here. The cardiac support system **205** can thus be fixed to the apparatus **100** to anchor the cardiac support system **205** in the blood vessel, for example in the aorta. The crown **110** ensures that the cardiac support system **205** is held in the aorta by a radial frictional connection of the apparatus with a wall section of the aorta.

FIG. 3 shows a schematic illustration of a cardiac support system **205** with an apparatus **100** for anchoring a cardiac support system **205** in a blood vessel according to a design example. The figure shows a side view of the cardiac support

system **205** with the apparatus **100**, whereby the apparatus **100** is configured in two parts. One part of the apparatus **100** comprises the crown **110** which is connected to the fixing means **105** by the connection means **115**, and another part of the apparatus **100** consists of the arching device **210**. The apparatus **100** is in an unfolded state corresponding to the anchoring state.

According to the design example shown here, the arching device **210** comprises an arch fixing device **305** for fixing the arching device **210** to the cardiac support system **205** and is not connected to the crown **110**. As shown here as an example, the arch fixing device **305** can fix the arching device **210** to the cardiac support system **205** by means of a form-locking and force-locking connection.

The insertion of the cardiac support system **205**, such as the heart pump shown here as an example, can preferably be performed in a minimally invasive manner through a human leg artery, the femoral artery. The insertion diameter available for the components of the apparatus **100** can likewise be limited by the maximum diameter of the femoral artery in the region of the implantation site or of other arteries over the course (for example the iliac artery, etc.) or even, for example, by the tortuosity or the degree of calcification of the arterial flow path. The cardiac support system **205** and the apparatus **100**, which can be brought into the body in this way, can consequently be limited in terms of diameter and overall length. In the case of a heart pump as the cardiac support system **205**, a high speed of the motor and the pump wheel of the heart pump can be set to nonetheless achieve significant support of the heart. The miniaturization of these components of the heart pump in combination with the high speed can lead to a decrease in the efficiency of the electric motor and an increase in the surface temperature as a result of the small heat-dissipating surface. If the apparatus **100** comprises an arching device **120** and is designed in one piece according to one embodiment as shown in the previous FIG. 2, the apparatus can fit snugly against the tube-shaped components of the cardiac support system **205** in the insertion state in order to keep the diameter as small as possible. The folded one-piece apparatus **100** can extend over the entire length of the motor-coupling-pump unit of the cardiac support system **205**. In this case, a two-part apparatus such as the design example shown in the present FIG. 3 is advantageous: A two-part design of the apparatus **100** and with it also a two-part anchoring of the cardiac support system **205** in the body makes it possible to save radial installation space in the region of the motor and impeller of the heart pump, for example in the region **310** of the cardiac support system **205** identified here. The installation space gained in this way in the region **310** can be used to enlarge the pump rotor of the cardiac support system **205**, for example, and thus reduce the speed in order to improve the efficiency. Or the diameter of the diameter of the coupling between the motor and the pump rotor of the cardiac support system **205** can be increased in the region **310** in order to be able to transmit a higher torque, or to save overall length of the coupling of the cardiac support system **205**. The region **310** can also be used to attach flow guiding geometries, for example guide vanes, to the housing of the cardiac support system **205** in order to thus increase the efficiency and to calm the blood flow.

FIG. 4 shows a schematic illustration of an insertion state of an apparatus **100** for anchoring a cardiac support system **205** in a blood vessel according to a design example in a side view. The apparatus **100** here comprises a sleeve **405**, into which the cardiac support system **205** is inserted. The cardiac support system **205** is also connected to the appa-

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ratus 100 by means of the fixing means 105, and the connection means 115 connects the crown 110 to the fixing means 105. The crown 110 is connected to the arching device 210.

According to the design example shown here, the apparatus 100 is cylindrical in the insertion state. For this purpose, the connection means 115, the crown 110 and the arching device 210 are folded together in a cylindrical manner; all of the components of the apparatus rest against the cardiac support system 205 and/or one respective other component of the apparatus 100.

The unfolding element, of which the crown 110 is configured, is optionally made of a shape memory material, for example Nitinol. Other components of the apparatus, such as the arching device 210 and the connection means 115, can likewise be made of a shape memory material. If, for example, the crown 110, the connection means 115, and the arching device 210 are cut out of a Nitinol tube, said components can be folded to their original cylindrical shape, the tube geometry, for the implantation process, due to the pseudoelastic properties of the shape memory material. The small installation space thus makes catheter-supported minimally invasive implantation possible. If the apparatus 100 comprises a sleeve 405 as in this case, the connection means 115, the crown 110 and the arching device 210 can be held down by the sleeve 405 and thus additionally or alternatively prevented from unfolding. In the illustration shown here, the aforementioned components of the apparatus 100, the crown 110, the connection means 115 and the arching device 210, are accordingly shown in the folded state inside the sleeve 405.

If the apparatus 100 comprises the sleeve 405 as according to a design example, the sleeve 405 is movable relative to the crown 110. The sleeve 405 is furthermore designed to enclose at least the crown 110 in the insertion state and to release at least the crown 110 to initiate the transition into the anchoring state. The sleeve 405 can thus be used to hold down the crown 110 and the arching device 210 in particular during the implantation process, so that these components do not unfold and thus do not stand up. For this purpose, the sleeve 405, also referred to as the release sheath, is pushed into a catheter over the other components of the apparatus 100 when the cardiac support system 205 connected to the fixing means 105 of the apparatus 100 is loaded into a catheter. As soon as the final position of the cardiac support system 205, e.g. in front of the heart valve, is reached, the sleeve 405 is pulled back, so that the crown 110 and the arching device 210 can unfold. Until then, the surgeon can reversibly determine the axial position, the rotational alignment and the position of the suction hose of the cardiac support system 205 in the ventricle. The sleeve 405 can furthermore be pulled back incrementally, so that there is a slow, gradual release of the arching device 210 and the crown 110 during the transition from the insertion state into the anchoring state.

FIG. 5 shows a schematic illustration of an insertion state of an apparatus 100 for anchoring a cardiac support system 205 in a blood vessel according to a design example. The figure shows a sectional view of the side view shown in the previous FIG. 4. The arching device 210 and the crown 110 are therefore in the loaded state and, together with the connection means and the fixing means 105, are enclosed by the sleeve 405, into which the cardiac support system 205 fixed to the apparatus 100 is inserted. The loaded state shown here makes it difficult to distinguish the components of the apparatus 100 inside the sleeve 405 from the cardiac support system 205, because they are in close contact with

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the cardiac support system 205. This illustrates the possibility of folding the apparatus 100 in a such a way that the cardiac support system 205 fixed to the apparatus direction 100 can be inserted and implanted in a minimally invasive manner.

FIG. 6 shows a schematic illustration of an apparatus 100 for anchoring a cardiac support system in a blood vessel having an arching device 210 according to a design example. The figure shows a side view of the apparatus 100 in the unfolded state in a one-piece design. The fixing means 105 comprises the configuration and the recesses for form-locking engagement in an element disposed on the cardiac support system by means of a clip connection. The connection means 115 here consists of four flexure struts 130 respectively connected at one end to the fixing means 105. The crown 110 consists of a plurality of unfolding elements 120 coupled to one another, the respective unfolding rods of which form rhombs. The respective other end of the four flexure struts 130 of the connection means 115 is connected to a connection of two adjacent unfolding elements 120. The arching device 210 is connected to the crown 110 at the end of the crown 110 opposite to the fixing means 105. The arching device 210 here comprises three feet 215. The feet 215 are respectively initially bent in a trough-shaped manner in the direction of the longitudinal axis of the apparatus 100 in one of their sections adjacent to the connection to the crown 110, and then form a semicircular arc in a section longer than half the length of each foot 215.

The crown 110 shown here can have an outer diameter in the unfolded state for anchoring the cardiac support system that is slightly larger than a human aorta, for example, so as to enable a uniform frictional connection to the blood vessel in the form of said aorta; the outer diameter in this case can be 20-30 mm, for example. The pressing force of the crown 110 and with it also the apparatus 100 is produced by the webs in the form of the unfolding elements 120, which are inclined at an angle of 20°-30°, in particular approx. 25°. The crown 110 advantageously has a conical shape with a 5°-10° angle which can ensure a continuous tangential connection to the release sheath in the form of the sleeve during the release process, i.e. during the transition from the insertion state into the anchoring state, and consequently a controlled release behavior, i.e. a controlled unfolding of the crown 110 and the arching device 210. The crown 110 has a length of 10-15 mm, in particular 13 mm. The fixing means 105 enables a connection to the cardiac support system, for example to a heart pump, via a connection to the motor housing of the cardiac support system. The rhomb shape of the unfolding elements 120 shown here represents a standard cross-section for vascular stents and makes a laser cut from a tube possible. The feet 215 of the arching device 210 respectively have an axial length of 20-30 mm, in particular 24 mm. At the front, i.e. at the end of the feet 215 opposite to the crown, the arching device 210 has a rotation diameter of 20-30 mm, in particular 23 mm, adapted to a human aortic valve. The feet 215 furthermore have an atraumatic shape, so that the aortic wall is not injured when the feet 215 are inserted and unfolded and the feet 215 can slide automatically into the cusps of the aortic valve. This anchoring shape of the feet 215 is advantageous, because the peanut shape of the cusps provides a particularly advantageous balance between contact surface and torsional rigidity. The indentation of the arching device 210 at the beginning of the feet 215, i.e. at the distal end connected to the crown 110, is designed such that, in the event of a partial release during the transition from the insertion state into the anchoring state, i.e. when the crown 110 is still in the crimped state, the target

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outer diameter of the arching device **210**, which corresponds to the rotation diameter, has already been reached.

The frame of the apparatus **100** consisting of the fixing means **105**, the connection means **115**, the crown **110** and the arching device **210** can, for example, be produced by means of the production method presented here using a shape memory material. For this purpose, an elastic material, preferably Nitinol or another shape memory alloy, is used for the frame. A tube geometry having the desired wall thickness of the later construction elements, the crown **110**, the feet **215** of the arching device **210**, the connection means **115** and the fixing means **105**, is a suitable semi-finished product for processing. The construction elements are realized using a method for material removal, preferably laser cutting, by removing pipe volume at the not needed locations. Punching and erosion methods or machining are alternatively possible as well. The laser-cut contour can now be brought into the desired shape, for example the shape shown here, as a part of a heat treatment, for example at a temperature above 500° C. The embossing process is a plastic deformation without the occurrence of material failure. The shape embossed in this way is then set automatically as soon as the transformation temperature is exceeded. This can be set via the alloy ratio and, in the application described, can be below body temperature, preferably between 0° C. and 10° C. To be able to serve patient groups having different sizes, the Nitinol frame of the apparatus **100** formed in this way can be made available in different sizes. For this purpose, the fixing of the apparatus **100** with the cardiac support system can, for example, take place right before loading the implantation device with which the cardiac support system connected to the apparatus **100** can be inserted. This allows the correct, patient-specific size of the crown **110** and the feet **215** to be selected shortly before implantation. When the cardiac support system is used for a longer period of time, the Nitinol structure of the frame allows it to grow together with the aorta, as the body's own tissue covers the biocompatible material.

FIG. 7 shows a schematic illustration of a foot **215** of an arching device according to a design example. The figure shows a plan view of the foot **215** of the arching device of the apparatus. The foot shown here corresponds to the uppermost foot **215** in the preceding FIG. 6, with the described arc-shaped short indentation followed by a semi-circular bulge, wherein the bulge does not represent an exact semicircle, but rather a flatter semicircular arc section. The arc shape of the foot can correspond to the form of a section of a human heart valve, for example a cusp of the aortic valve. During the transition of the apparatus from the insertion state into the anchoring state, the foot **215** can unfold first. The foot **215** can then be positioned to align the cardiac support system in the blood vessel, for example behind the heart valve in the cusps of the aortic valve. If the desired positioning is not immediately successful, the feet **215** can also be folded again, even repeatedly, for example using the sleeve, by sliding the sleeve forward again, i.e. over the feet **215**, and then sliding it back to reposition the feet **215** and thus allowing the feet **215** to unfold.

FIG. 8 shows a schematic illustration of a part of a crown **110** of an apparatus for anchoring a cardiac support system in a blood vessel according to a design example. As a section of the crown **110**, the figure shows a plan view onto three unfolding elements **120** of the crown **110**, which are unfolded into a rhomb shape, in a conical configuration of the crown **110**. The sectional contour of the crown **110** shown here corresponds to the not perfectly circular aortic anatomy of the human aorta.

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During the transition of the apparatus from the insertion state into the anchoring state, for example, the arching device can be unfolded first and then the crown **110**. The shape of the crown in the unfolded state, as shown here, can be such that a uniform force-locking connection is created between the crown **110** and the aorta. This connection allows the cardiac support system to be held in position, and the resulting anchoring of the cardiac support system in the blood vessel prevents dislocation of the cardiac support system during operation. For this purpose, the outer diameter of the crown **110** in the unfolded state of the crown **110** is slightly larger than the inner diameter of the aorta.

FIG. 9 shows a schematic illustration of a fixing means **105** of an apparatus for anchoring a cardiac support system **205** in a blood vessel according to a design example. The plan view shows the section of the cardiac support system **205** connected to the fixing means **105**. According to a design example, the fixing means **105** is designed to fix the apparatus in a form-locking and/or force-locking manner to the cardiac support system **205**. The fixing can also be realized in a material-locking manner. The attachment by means of the fixing means **105** can be configured such that the apparatus cannot be detached from the cardiac support system **205** during operation of the cardiac support system **205** and the attachment can remain stable over the operating time of the cardiac support system **205**. The fixing means **105** can further be configured to absorb forces occurring during operation of the cardiac support system **205** or during the surgical procedure. The fixing means **105** can be designed to support the positioning of the cardiac support system in the blood vessel. The relative position of the cardiac support system **205** and the fixing means can be adjustable via a rotational movement in both directions and can be designed to be incrementally variable over 360°, and the fixing means **105** and the cardiac support system **205** can consequently be movable relative to one another prior to the form-locking fixing.

For form-locking and/or force-locking fixing of the cardiac support system **205** according to a design example, the fixing means **105** and the cardiac support system **205** can have features that allow the cardiac support system to snap into the apparatus and thus enable an anchoring of the cardiac support system **205** to the apparatus. For this purpose, the cardiac support system **205** and/or the fixing means **105** can comprise at least one disposed element **905** for a form-locking connection of the apparatus and the cardiac support system **205** by means of the fixing means **105**. In FIG. 9 here, as an example, the element is disposed on the cardiac support system, is round and has an undercut in cross-section for engagement with the clip connection of the fixing means **105** shown here, which is formed to correspond to the element **905**. The clip connection here comprises two locking springs **910** in order to be able to easily connect the fixing means to the cardiac support system, and to achieve additional fixing by means of the locking springs **910** after the engagement of the form-locking element. A latching of the fixing means to the cardiac support system **205**, which is configured here in the form of the locking springs **910**, can be realized irrespective of the rotational offset of the disposed element **905** relative to the corresponding recess.

The following FIGS. 10 to 17 show various design examples for form-locking and/or force-locking fixing of the cardiac support system **205** to the apparatus.

FIG. 10 shows a schematic illustration of a fixing means **105** of an apparatus for anchoring a cardiac support system **205** in a blood vessel according to a design example. The plan view shows a section of the cardiac support system **205**

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connected to a section of the fixing means **105**. A round element **905** is disposed on the cardiac support system, and a section of the fixing means comprises a recess corresponding to the element **905** for form-locking engagement of the fixing means **105** to securely fix the cardiac support system **205** to the apparatus.

FIGS. **11a** to **11j** respectively show a schematic illustration of a design of a form-locking fixing of an apparatus for anchoring a cardiac support system in a blood vessel according to a design example. The figure shows a possible configuration of the disposed element shown in the two preceding FIGS. **9** and **10** for form-locking connection of the cardiac support system to the apparatus by means of the engagement of the disposed element and a corresponding recess or configuration as a counterpart. In the manner of a positive and negative design, such a disposing element for form-locking fixing can be realized on both the cardiac support system, for example the housing of the pump, and on the connection mechanism of the apparatus, i.e. on the fixing means. FIGS. **11a** to **11e** show the various shapes that the element for form-locking fixing can assume as a result of the configuration in a schematic plan view, and FIGS. **11f** to **11j** show the various cross-sections that such an element can have. The element can be shaped as a circle according to the shape shown in FIG. **11a**, or have an oval shape according to the shape shown in FIG. **11b**. According to FIG. **11c**, the element can be a triangle, or a polygon as shown in FIG. **11d**. The element can also be shaped as a star as shown in FIG. **11e**. The element can have a rectangular cross-section as shown in FIG. **11f**, or the flattening of the cross-section shown in FIG. **11g**, the rounding of the cross-section shown in FIG. **11h**, the semicircle of the cross-section shown in FIG. **11i**, or the undercut of the cross-section shown in FIG. **11j**. Combinations of the shown shapes can also be used to shape the element.

FIG. **12** shows a schematic illustration of a force-locking fixing of an apparatus **100** for anchoring a cardiac support system **205** in a blood vessel to a cardiac support system **205** according to a design example. The plan view shows a section of the cardiac support system **205** and the fixed apparatus **100**. The apparatus **100** comprises a fixing means having a plurality of rhomb-shaped fixing elements, which are configured to prevent the heart support system **205** and the apparatus **100** from shifting relative to one another. For this purpose, according to the design example shown here, the fixing elements rest on a section of the cardiac support system **205** and clamp the apparatus against the cardiac support system **205** to fix the apparatus in a force-locking manner to the cardiac support system **205**.

FIG. **13** shows a schematic illustration of a part of a fixing means **105** of an apparatus for anchoring a cardiac support system in a blood vessel according to a design example. The figure shows a plan view onto a connection of the apparatus to the cardiac support system, which is realized in a form-locking manner as a bayonet connection. In this case, the fixing means **105** comprises a bayonet connection in the form of a hook.

FIG. **14** shows a schematic illustration of a part of a fixing means of an apparatus for anchoring a cardiac support system **205** in a blood vessel according to a design example in cross-section. The fixing means **105** here is configured for form-locking fixing, the cardiac support system comprises a corresponding configuration. The fixing means and the cardiac support system respectively comprise a groove for fixing.

FIG. **15** shows a schematic illustration of a part of a fixing means **105** of an apparatus for anchoring a cardiac support

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system **205** in a blood vessel according to a design example. The figure shows a section of the fixing means **105** and a section of the cardiac support system **205** as examples, whereby the sections engage in one another in a form-locking manner. The plan view shows the form-locking connection of the fixing means **105** to the cardiac support system via an engagement of the fixing means **105** in two hooks of the cardiac support system **205**.

FIG. **16** shows a schematic illustration of a part of a fixing means of an apparatus for anchoring a cardiac support system **205** in a blood vessel according to a design example. The figure shows a cross-section of a force-locking connection of the cardiac support system **205** to the fixing means **105**, which is realized here as a spring clamp via a spring connection using two springs **1605** between the heart support system **205** and the fixation device **105**.

FIG. **17** shows a schematic illustration of a part of a fixing means of an apparatus for anchoring a cardiac support system **205** in a blood vessel according to a design example. The figure shows a cross-section of a press connection as the force-locking connection of the cardiac support system **205** to the fixing means **105**. The assembly of the press connection shown here can be produced with the aid of a pressing force, or with the aid of a temperature difference between the cardiac support system **205** and the fixing means **105**, i.e. by press joining with the aid of a temperature difference between the fixing means **105** and the cardiac support system **205** and the pressure that occurs during the subsequent temperature compensation.

FIGS. **18a** to **18c** each show a schematic illustration of an apparatus **100** for anchoring a cardiac support system in a blood vessel according to a design example, whereby each figure shows a different situation for the apparatus **100**. FIG. **18a** shows a plan view of the unfolded apparatus **100**, as already described with reference to FIG. **1**. FIG. **18b** shows a section of the apparatus shown in FIG. **18a** in the sleeve **405** in the insertion state, as already described with reference to FIG. **4**. FIG. **18c** shows a side view of the apparatus **100**. The situations shown here are examples of the space required by the apparatus **100** in the insertion state depending on the configuration of the apparatus **100**. The following FIGS. **19a** to **19c** show the situations described here according to an alternative configuration of the crown **110** of the apparatus **100** according to a design example.

FIG. **18a** shows a side view of the apparatus **100** according to a design example. The apparatus **100** comprises the fixing means **105**, the connection means **115** and the crown **110**, whereby the crown **110** is made of a plurality of unfolding elements **120** coupled to one another. Each of the unfolding elements **120** comprises two unfolding rods connected at their ends, which are rhomb-shaped in the unfolding state shown here. The connection means **115** consists of four flexure struts **130**, wherein a first end of each flexure strut **130** is fastened to the fixing means **105** and wherein a second end of each flexure strut **130** is fastened to a connection between two adjacent unfolding elements **120**.

FIG. **18b** shows the apparatus **100** described in FIG. **18a** in the insertion state. The crown **110** here is folded for insertion and inserted into the sleeve **405**. Due to the described configuration of the crown **110** and the type of fastening of the flexure struts **130**, the apparatus **100**, also referred to as the anchor, has a comparatively high axial space requirement, for example in comparison to the design example of the apparatus **100** shown in FIG. **19b**. According to this design example, the flexure struts **130** are not completely folded in under or between the unfolding elements of the crown **110**; rather, they require a certain amount

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of axial space in the insertion state, which is shown in this figure by the gap between the loaded crown 110 and the fixing means 105.

FIG. 18c schematically shows an illustration of the apparatus 100 in the finished state. The apparatus 100 accordingly does not have the cylindrical shape of the insertion state or the unfolded shape of the anchoring state. The finished state shown here shows an example of the space required by the connection and shape of the crown 110 and the flexure struts 130 of the connection means.

FIG. 19a to 19c each show a schematic illustration of an apparatus 100 for anchoring a cardiac support system in a blood vessel according to a design example. The situations shown in FIG. 19a to 19c each show the apparatus 100 as it is described with reference to FIG. 19a. Shown is a configuration of the crown 110 and the connection means 115, with which the apparatus 100 in the loaded state, as in the folded state corresponding to the insertion state, has a comparatively low axial space requirement. This can be advantageous, for example, if the apparatus is used in conjunction with a cardiac support system, for which additional installation space in this region is advantageous, for example for operating a motor of a heart pump. The saving of axial space can be achieved by a modified geometry of the crown 110 and the connection means 115. The shape of the crown 110, the fastening of the flexure struts 130, as well as the long flexure struts 130 and their change in direction can largely prevent the anchor, i.e. the apparatus 100, from taking up unnecessary installation space during loading. According to the design example shown here, the crown 110 furthermore unfolds at the place at which it is also located in the loaded state.

FIG. 19a shows the apparatus 100 in side view in the unfolded state, whereby the apparatus 100 according to this design example has a low axial space requirement. The crown 110 consists of the at least one unfolding element having a plurality of loops disposed in a meandering manner, as shown in FIG. 19c. The crown 110 is fastened to the fixing means 105.

FIG. 19b shows the apparatus 100 described in FIG. 19a in the insertion state. The crown 110 here is crimped for insertion and inserted into the sleeve 405. Due to the configuration of the crown 110 and the type of fastening of the flexure struts 130, the apparatus 100, also referred to as the anchor, has a comparatively low axial space requirement, for example in comparison to the design example of the apparatus 100 shown in FIG. 18b. According to this design example, the flexure struts 130 are folded in between the unfolding elements of the crown 110; they require only a small amount of axial space in the insertion state, which is shown in this figure by the small gap between the crimped crown 110 and the fixing means 105.

FIG. 19c schematically shows an illustration of the apparatus 100 in the finished state as a further situation. This figure clearly shows the described configurations of the unfolding element 120 with a plurality of meander-like loops and the described connection of the flexure struts 130 to the crown 110. The crown 110 can thus be made of only one single unfolding element 120. The flexure struts 130 of the connection means are disposed inside the meander-like loops, whereby a first end of each flexure strut 130 is fastened to the fixing means 105 and whereby a second end of the flexure strut 130 is fastened to an end of the crown 110 facing away from the fixing means 105. This connection of the flexure struts 130 to the crown 110 is particularly advantageous in terms of the axial space requirement of the

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apparatus 100 in the insertion state, as shown in the previous situation of the apparatus 100 in FIG. 19b.

FIG. 20 shows a flow diagram of a method 2000 for operating an apparatus for anchoring a cardiac support system in a blood vessel according to a design example. The method 2000 comprises at least one step 2001 of unfolding the unfolding element of the crown of the apparatus during the transition from the insertion state into the anchoring state to increase the diameter of the crown.

FIG. 21 shows a flow diagram of a production method 2100 for producing an apparatus for anchoring a cardiac support system in a blood vessel according to a design example. The apparatus can assume an insertion state for inserting the cardiac support system into the blood vessel, and the apparatus can assume an anchoring state for anchoring the cardiac support system in the blood vessel. The production method 2100 of the apparatus comprises at least one step 2101 for providing, one step 2103 for forming and one step 2105 for heat treating. In the providing step 2101, a semi-finished product made of a shape memory material is provided. In the forming step 2103, a fixing means for fixing the apparatus to the cardiac support system is formed. The step also includes forming a crown from at least one unfolding element, whereby the unfolding element is designed to unfold during a transition from the insertion state into the anchoring state in order to increase the diameter of the crown to anchor the apparatus in the blood vessel. A connection means is furthermore configured to connect the crown to the connection means. The fixing means, the crown and the connection means are formed from the semi-finished product provided in Step 2101. In the heat-treating step 2105, the fixing means, the crown and the connection means are heat-treated to emboss the shape of the anchoring state.

If a design example includes an “and/or” conjunction between a first feature and a second feature, this should be read to mean that the design example according to one embodiment comprises both the first feature and the second feature and, according to another embodiment, comprises either only the first feature or only the second feature.

In summary, the following preferred features of the invention should in particular be noted: An apparatus 100 for anchoring a cardiac support system in a blood vessel can assume an insertion state for inserting the cardiac support system into the blood vessel and an anchoring state in order to anchoring state the cardiac support system into the blood vessel. The apparatus 100 comprises at least one fixing means 105 for fixing the apparatus 100 to the cardiac support system 205, a crown 110 and a connection means 115. The crown 110 consists of at least one unfolding element 120. The unfolding element 120 is designed to unfold during a transition from the insertion state into the anchoring state in order to increase the diameter of the crown 110 to anchor the apparatus 100 in the blood vessel. The connection means 115 is designed to connect the crown 110 to the fixing means 105.

The invention in particular relates to the aspects specified in the following clauses:

1. Apparatus (100) for anchoring a cardiac support system (205) in a blood vessel, wherein the apparatus (100) can assume an insertion state for inserting the cardiac support system (205) into the blood vessel, and an anchoring state for anchoring the cardiac support system (205) in the blood vessel, and wherein the apparatus (100) has the following features:
 - a fixing means (105) for fixing the apparatus (100) to the cardiac support system (205);

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- a crown (110) that consists of at least one unfolding element (120), wherein the unfolding element (120) is designed to unfold during a transition from the insertion state into the anchoring state in order to increase the diameter of the crown (110) to anchor the apparatus (100) in the blood vessel; and
 a connection means (115) which is configured to connect the crown (110) to the fixing means (105).
2. Apparatus (100) according to Clause 1, wherein at least the unfolding element (120) is made of a shape memory material.
 3. Apparatus (100) according to any one of the preceding clauses, having an arching device (210) having at least one foot (215), wherein the arching device (210) is designed to unfold during the transition from the insertion state into the anchoring state for positioning the at least one foot (215) in the blood vessel, and wherein the arching device (210) is connected to the crown (110) or wherein the arching device (210) comprises an arch fixing device (305) for fixing the arching device (210) to the cardiac support system (205).
 4. Apparatus (100) according to Clause 3, wherein the arching device (210) comprises three feet (215), in particular wherein, for positioning the feet (215), said feet (215) are formed in a respective cusp of a heart valve.
 5. Apparatus (100) according to any one of the preceding clauses, wherein the apparatus (100) is cylindrical in the insertion state.
 6. Apparatus (100) according to any one of the preceding clauses, wherein the unfolding element (120) has an inclined position relative to the longitudinal axis of the apparatus (100) in the anchoring state.
 7. Apparatus (100) according to any one of the preceding clauses, wherein the crown (110) comprises a plurality of unfolding elements (120) coupled to one another, wherein each of the unfolding elements (120) comprises two unfolding rods (125) connected at their ends, wherein the distance between the two unfolding rods (125) is smaller in the insertion state than in the anchoring state.
 8. Apparatus (100) according to any one of Clauses 1 to 6, wherein the unfolding element (120) comprises a plurality of loops arranged in a meandering manner, wherein the distance between the loops is smaller in the insertion state than in the anchoring state.
 9. Apparatus (100) according to any one of the preceding clauses, wherein the connection means (115) comprises at least one flexure strut (130), wherein the flexure strut (130) is designed to open during the transition from the insertion state into the anchoring state to allow the crown (110) to unfold.
 10. Apparatus (100) according to Clause 9, wherein a first end of the flexure strut (130) is fastened to the fixing means (105) and wherein a second end of the flexure strut (130) is fastened to a connection between two adjacent unfolding elements (120) or to an end of the crown (110) facing away from the fixing means (105).
 11. Apparatus (100) according to any one of the preceding clauses, wherein the fixing means (105) is designed to fix the apparatus (100) in a form-locking and/or force-locking manner to the cardiac support system (205).
 12. Apparatus (100) according to any one of the preceding clauses with a sleeve (405), wherein the sleeve (405) is movable relative to the crown (110) and wherein the sleeve (405) is designed to enclose at least the crown

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- (110) in the insertion state and to release at least the crown to initiate the transition into the anchoring state.
13. Method (2000) for operating an apparatus (100) according to any one of the preceding clauses, wherein the method (2000) comprises at least the following step:
 unfolding (2001) the unfolding element (120) of the crown (110) of the apparatus (100) during the transition from the insertion state into the anchoring state to increase the diameter of the crown (110).
 14. Production method (2100) for producing an apparatus (100) for anchoring a cardiac support system (205) in a blood vessel, wherein the apparatus (100) can assume an insertion state for inserting the cardiac support system (205) into the blood vessel, and wherein the apparatus (100) can assume an anchoring state for anchoring the cardiac support system (205) in the blood vessel, wherein the production method comprises at least the following step:
 providing (2101) a semi-finished product made of a shape memory material;
 forming (2103) a fixing means (105) for fixing the apparatus (100) to the cardiac support system (205) and forming a crown (110) from at least one unfolding element (120), wherein the unfolding element (120) is configured to unfold during a transition from the insertion state into the anchoring state to increase the diameter of the crown (110) in order to anchor the apparatus (100) in the blood vessel, and forming a connection means (115), that is shaped to connect the crown (110) to the connection means, wherein the fixing means, the crown (110) and the connection means (115) are made of the semi-finished product; and heat treating (2105) the fixing means, the crown (110) and the connection means (115) in order to emboss the shape of the anchoring state.
 15. Cardiac support system (205) having an apparatus (100) according to any one of the preceding clauses.
 The invention claimed is:
 1. An apparatus for anchoring a cardiac support system in a blood vessel, the apparatus comprising:
 a fixing component configured to fix to the cardiac support system;
 a ring-shaped crown, wherein a diameter of the ring-shaped crown is configured to increase during a transition of the apparatus from an insertion state into an anchoring state so as to anchor the cardiac support system in the blood vessel, wherein the apparatus is configured to assume the insertion state for inserting the cardiac support system into the blood vessel and to assume the anchoring state for anchoring the cardiac support system in the blood vessel;
 a sleeve being movably displaceable relative to the crown during the transition from the insertion state into the anchoring state so as to allow the crown to unfold; and
 a connection component configured to connect the crown to the fixing component, the connection component comprising a plurality of flexure struts configured to unfold in response to displacement of the sleeve, wherein each flexure strut of the plurality of flexure struts comprises a first end coupled to the fixing component,
 wherein the ring-shaped crown comprises a plurality of unfolding elements configured to unfold during the transition of the apparatus from the insertion state into the anchoring state so as to increase the diameter of the crown,

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wherein each unfolding element comprises two unfolding rods connected to one another at their ends,
 wherein the plurality of unfolding elements connect to one another along a circular path, wherein a central section of each unfolding rod of each unfolding element is connected at a connection point to a central section of an adjacent unfolding rod of an adjacent unfolding element,
 wherein each unfolding rod is disposed closer to each adjacent unfolding rod in the insertion state than in the anchoring state, wherein each of the plurality of flexure struts comprises a second end connected to the connection point of two unfolding rods of adjacent unfolding elements or to a connection at the ends of two unfolding rods of one of the unfolding elements at an end of the crown opposite the fixing component, and wherein the sleeve is configured to enclose the crown in the insertion state and is configured to release the crown so as to initiate the transition from the insertion state into the anchoring state.

2. The apparatus according to claim 1, wherein the unfolding elements are made of a shape memory material.

3. The apparatus according to claim 1, further comprising an arching device comprising at least one foot, wherein the arching device is configured to unfold during the transition from the insertion state into the anchoring state so as to position the at least one foot in the blood vessel, and wherein the arching device is connected to the crown or the arching device comprises an arch fixing device for fixing the arching device to the cardiac support system.

4. The apparatus according to claim 3, wherein the at least one foot comprises three feet, wherein each of the three feet are configured to be positioned in a cusp of a heart valve.

5. The apparatus according to claim 1, wherein the apparatus is cylindrical in the insertion state.

6. The apparatus according to claim 1, wherein the crown and the fixing component extend along a longitudinal axis, and wherein each of the plurality of unfolding elements is configured to be positioned in an inclined position relative to the longitudinal axis of the apparatus in the anchoring state.

7. The apparatus according to claim 1, wherein the fixing component is configured to fix the apparatus to the cardiac support system in a form-locking and/or force-locking manner.

8. A method for producing an apparatus for anchoring a cardiac support system in a blood vessel, the method comprising:

- providing a semi-finished product made of a shape memory material;
- forming a fixing component, a ring-shaped crown and a connection component from the semi-finished product; and
- heat treating the fixing component, the crown and the connection component in order to emboss a shape of the anchoring state;

wherein the fixing component is configured to fix to the cardiac support system,

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wherein a diameter of the ring-shaped crown is configured to increase during a transition of the apparatus from an insertion state into an anchoring state so as to anchor the cardiac support system in the blood vessel, wherein the apparatus is configured to assume the insertion state for inserting the cardiac support system into the blood vessel and to assume the anchoring state for anchoring the cardiac support system in the blood vessel,
 wherein the apparatus further comprises a sleeve being movably displaceable relative to the crown during the transition from the insertion state into the anchoring state so as to allow the crown to unfold,
 wherein the connection component is configured to connect the crown to the fixing component, the connection component comprising a plurality of flexure struts configured to unfold in response to displacement of the sleeve, wherein each flexure strut of the plurality of flexure struts comprises a first end coupled to the fixing component,
 wherein the ring-shaped crown comprises a plurality of unfolding elements configured to unfold during the transition of the apparatus from the insertion state into the anchoring state so as to increase the diameter of the crown,
 wherein each unfolding element comprises two unfolding rods connected to one another at their ends,
 wherein the plurality of unfolding elements connect to one another along a circular path, wherein a central section of each unfolding rod of each unfolding element is connected at a connection point to a central section of an adjacent unfolding rod of an adjacent unfolding element,
 wherein each unfolding rod is disposed closer to each adjacent unfolding rod in the insertion state than in the anchoring state, wherein each of the plurality of flexure struts comprises a second end connected to the connection point of two unfolding rods of adjacent unfolding elements or to a connection at the ends of two unfolding rods of one of the unfolding elements at an end of the crown opposite the fixing component, and
 wherein the sleeve is configured to enclose the crown in the insertion state and is configured to release the crown so as to initiate the transition from the insertion state into the anchoring state.

9. The method according to claim 8, wherein the apparatus further comprises an arching device comprising at least one foot, wherein the arching device is configured to unfold during the transition from the insertion state into the anchoring state so as to position the at least one foot in the blood vessel, and wherein the arching device is connected to the crown or the arching device comprises an arch fixing device for fixing the arching device to the cardiac support system.

10. The method according to claim 8, wherein the fixing component is configured to fix the apparatus to the cardiac support system in a form-locking and/or force-locking manner.

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